HARISH PAL

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CAREER OBJECTIVE

Software Engineering student skilled in Python, backend development, algorithms, and Generative AI, with experience building scalable LLM-powered and agentic applications.

EDUCATION

B. Tech in CSE-AI	Noida Institute of Engineering Technology	2023-Present	9.02 CGPA
Class XII, CBSE Board	ATL School	2021–2022	90.00%
Class X, CBSE Board	ATL School	2019–2020	93.17%

PROJECTS

R.AI — Autonomous Revenue Recovery Agent GitHub

Python, FastAPI, Next.js, Docker, TypeScript, Razorpay APIs, PostgreSQL,

- Built an **agentic payment recovery platform** for 10K+ transactions, using LLM-driven diagnosis/recommendations with deterministic policy enforcement, human approval, idempotent execution, and provider-backed recovery workflows.
- Designed an **AI-vs-baseline evaluation framework** and outcome-observation pipeline to distinguish *recommended → executed → observed → recovered* states and quantify recovery performance.
- Implemented efficient semantic indexing and retrieval using Qdrant to support low-latency search across large collections of financial documents.
- Developed a full-stack **Revenue Intelligence dashboard** with analytics, recovery lifecycle tracking, audit trails, approval workflows, and deterministic demo/evaluation modes; maintained **66/66 backend tests passing** with frontend lint, type-check, and production build validation.

CodeAutopsy — Codebase Reverse Engineering & Architecture Analysis Platform GitHub

FastAPI, Tree-sitter, NetworkX, FAISS, sentence-transformers, Next.js, React Flow

- Built a backend platform that reverse-engineers GitHub repositories into architecture documentation, dependency graphs, call graphs, and searchable code insights.
- Designed a multi-stage analysis pipeline combining AST parsing, graph construction, semantic indexing, and architecture inference for large-scale codebases.
- Integrated deterministic static analysis with LLM-assisted reasoning to generate module summaries, detect architectural anti-patterns, identify bug-prone components, and produce explainable analysis reports.
- Developed an interactive React Flow dashboard for visualizing repository structure, dependency graphs, intermediate analysis artifacts, and code relationships.

SKILLS

- **Programming Languages**: C++, Java, Python, JavaScript, TypeScript
- **Core Computer Science:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Database Management Systems (DBMS), Computer Networks, System Design
- **Backend**: FastAPI, Node.js, Express.js, RESTful APIs, Authentication, API Design, Concurrency,
- **Frontend**: React.js, Next.js, Tailwind CSS
- **Databases**: MySQL, MongoDB, PostgreSQL
- **AI & Data**: Generative AI, RAG, LangChain, Agentic Systems, n8n, Embeddings, Semantic Retrieval, FAISS, Qdrant
- **Infrastructure & Tools**: Docker, Linux, Git/GitHub, AWS

INTERNSHIPS

Python Developer – LetsGrowMore JULY 2024–AUGUST 2024

- Developed Python automation scripts using **Pandas** to process datasets, improving internal data workflows.
- Implemented data validation and logging mechanisms to ensure reliable data processing and easier debugging.
- Assisted in building backend utilities using **Python/Django**, supporting API development and data handling tasks.

ACHIEVEMENTS

- **LeetCode**: 700+ DSA problems solved | Contest Rating: 1850+
- Top **10** finalist among 120+ teams in **HackO'Clock** National Hackathon.
- Published Research paper on SSRN: *"Predicting Growth-Induced Financial Collapse of SMEs Using Structural Cash-Flow Stress Modeling."*